

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Viva Voice Questions

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 –Viva Voice Questions	Assessment:	Assessment Tool 3– Viva Voice Questions
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	POOJA S	Reg. No.:	192524211
Section / Batch:		Date:	20/08/2026

Code of Conduct

I ____POOJA S(192524211)____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ____POOJA S____

A. Introduction to Statistical Inference

1. What is statistical inference?

Answer: Statistical inference is the process of using data collected from a sample to make conclusions, predictions, or decisions about a larger population.

2. Why is statistical inference important in data science?

Answer: Statistical inference is important in data science because it helps data scientists draw meaningful conclusions about a population based on sample data. It also helps in prediction, decision-making, and testing assumptions.

3. What is the difference between descriptive statistics and inferential statistics?

Answer: Descriptive statistics summarize and describe the collected data using measures such as mean, median, and standard deviation. Inferential statistics use sample data to make conclusions or predictions about a larger population.

4. What is a population in statistics?

Answer: A population is the complete set of all individuals, objects, or observations that are of interest in a statistical study.

5. What is a sample?

Answer: A sample is a smaller group or subset selected from a population for analysis.

6. Why do data scientists usually work with samples instead of complete populations?

Answer: Data scientists usually work with samples because collecting data from the entire population may be expensive, time-consuming, difficult, or sometimes impossible. A properly selected sample can provide useful information about the population.

7. What is a population parameter?

Answer: A population parameter is a numerical value that describes a characteristic of an entire population, such as the population mean, population proportion, or population variance.

8. What is a sample statistic?

Answer: A sample statistic is a numerical value calculated from sample data. It is used to estimate a corresponding population parameter.

9. Give a real-world example where statistical inference is used in industry.

Answer: An e-commerce company may select a sample of customers to determine whether a new website design increases the average amount spent by customers. The results from the sample are then used to make conclusions about all customers.

10. What are the two major activities involved in statistical inference?

Answer: The two major activities involved in statistical inference are estimation and hypothesis testing. Estimation is used to estimate population parameters, while hypothesis testing is used to test claims or assumptions about a population.

B. Frequentist Approach

11. What is the frequentist approach to statistical inference?

Answer: The frequentist approach is a method of statistical inference that interprets probability as the long-run frequency of an event occurring in repeated experiments or observations.

12. What is the basic idea behind frequentist statistics?

Answer: The basic idea is that population parameters are fixed but unknown, while sample data can vary due to random sampling. Statistical conclusions are based on the probability of observing the sample results.

13. How does the frequentist approach differ from the Bayesian approach?

Answer: In the frequentist approach, parameters are considered fixed and probability is based on long-run frequency. In the Bayesian approach, prior information and observed data are combined to update beliefs about unknown parameters.

14. In the frequentist approach, what is considered fixed: the parameter or the data?

Answer: In the frequentist approach, the population parameter is considered fixed but unknown, while the data is considered random because different samples can produce different results.

15. What does long-run frequency mean in frequentist statistics?

Answer: Long-run frequency refers to the proportion of times an event would occur if the same experiment were repeated a very large number of times under the same conditions.

16. Why is random sampling important in the frequentist approach?

Answer: Random sampling is important because it helps ensure that the sample represents the population and reduces selection bias. It also allows probability-based statistical methods to be applied correctly.

17. Can a population parameter be considered a random variable in the frequentist approach? Why?

Answer: No. In the frequentist approach, a population parameter is considered fixed but unknown. It does not change randomly; only the sample and sample statistics vary from one random sample to another.

18. Give an example of a frequentist problem in a data science application.

Answer: A data scientist may test whether a new machine-learning model has a significantly higher accuracy than an existing model. A hypothesis test and p-value can be used to determine whether the observed improvement is statistically significant.

C. Point Estimation

19. What is a point estimate?

Answer: A point estimate is a single numerical value calculated from sample data and used to estimate an unknown population parameter.

20. What is an estimator?

Answer: An estimator is a statistical formula, rule, or method used to calculate an estimate of an unknown population parameter from sample data.

21. What is the difference between an estimator and an estimate?

Answer: An estimator is the formula or method used to estimate a population parameter, while an estimate is the actual numerical value obtained after applying the estimator to a particular sample.

22. What is the point estimator for the population mean?

Answer: The point estimator for the population mean is the sample mean, denoted by $\bar{x}$.

23. How do you estimate a population proportion from a sample?

Answer: A population proportion is estimated using the sample proportion. It is calculated by dividing the number of successes by the total sample size.

24. What is the point estimate of population variance?

Answer: The population variance is commonly estimated using the sample variance, denoted by s^2 .

25. Why is the sample mean commonly used to estimate the population mean?

Answer: The sample mean is commonly used because it is easy to calculate, uses all observations in the sample, and is an unbiased estimator of the population mean under random sampling.

26. What properties should a good estimator have?

Answer: A good estimator should generally have properties such as unbiasedness, consistency, efficiency, and low variance.

27. What is an unbiased estimator?

Answer: An unbiased estimator is an estimator whose expected value is equal to the true population parameter being estimated.

28. What is meant by the bias of an estimator?

Answer: Bias is the difference between the expected value of an estimator and the true value of the population parameter.

29. What is the difference between bias and variance of an estimator?

Answer: Bias measures how far the average estimated value is from the true population parameter. Variance measures how much the estimator changes or varies from one sample to another.

30. Why can two different samples produce different point estimates for the same population parameter?

Answer: Two different samples may contain different observations due to random sampling. Therefore, the calculated sample statistics and point estimates can vary even though they are estimating the same population parameter.

D. Variability in Estimates

31. Why does a point estimate have uncertainty?

Answer: A point estimate has uncertainty because it is calculated from a sample rather than the entire population. Different random samples can produce different estimates.

32. What is the sampling distribution of an estimator?

Answer: The sampling distribution of an estimator is the probability distribution of the estimator obtained from all possible random samples of the same size from a population.

33. What is the standard error?

Answer: Standard error is the standard deviation of the sampling distribution of a statistic. It measures how much a sample estimate is expected to vary from sample to sample.

34. How is standard error different from standard deviation?

Answer: Standard deviation measures the variability of individual observations in a dataset. Standard error measures the variability or uncertainty of a sample statistic, such as the sample mean.

35. How does increasing the sample size affect the standard error of the sample mean?

Answer: Increasing the sample size decreases the standard error of the sample mean. A larger sample generally provides a more precise estimate of the population mean.

36. Why does variability in an estimate decrease when the sample size increases?

Answer: As the sample size increases, random differences between samples tend to average out. Therefore, larger samples generally produce estimates with less variability and greater precision.

37. What is the relationship between standard error and confidence intervals?

Answer: Standard error is used to calculate the margin of error in a confidence interval. A smaller standard error generally produces a narrower confidence interval, while a larger standard error produces a wider confidence interval.

38. If two samples have different sample means, does it necessarily mean that one sample is incorrect? Explain.

Answer: No. Different random samples can naturally produce different sample means because of sampling variability. Both samples may be correct and still provide different estimates of the same population mean.

E. Confidence Intervals

39. What is a confidence interval?

Answer: A confidence interval is a range of values calculated from sample data that is used to estimate an unknown population parameter with a specified level of confidence.

40. Why is a confidence interval more informative than a point estimate?

Answer: A confidence interval is more informative because it provides a range of plausible values for the population parameter and shows the uncertainty or precision associated with the estimate.

41. What does a 95% confidence level mean in the frequentist approach?

Answer: A 95% confidence level means that if the same sampling process were repeated many times and a confidence interval were calculated each time, approximately 95% of those intervals would contain the true population parameter.

42. What is the difference between a confidence level and a confidence interval?

Answer: The confidence level is the percentage used to describe the reliability of the confidence interval, such as 95%. A confidence interval is the actual range of values calculated from the sample data.

43. What factors determine the width of a confidence interval?

Answer: The width of a confidence interval depends mainly on the sample size, variability in the data, and confidence level. A larger sample size generally makes the interval narrower, while greater variability or a higher confidence level makes it wider.

44. How does increasing the sample size affect the width of a confidence interval?

Answer: Increasing the sample size generally decreases the width of the confidence interval because the estimate becomes more precise and the standard error becomes smaller.

45. How does increasing the confidence level from 95% to 99% affect the confidence interval?

Answer: Increasing the confidence level from 95% to 99% makes the confidence interval wider because a larger range is required to achieve greater confidence that the interval contains the true population parameter.

46. What is the difference between a z-confidence interval and a t-confidence interval?

Answer: A z-confidence interval uses the standard normal distribution and is generally used when the population standard deviation is known or the sample size is large. A t-confidence interval uses the t-distribution and is generally used when the population standard deviation is unknown, especially for small samples.

47. When would you use the t-distribution instead of the normal distribution for estimating a population mean?

Answer: The t-distribution is used when estimating a population mean if the population standard deviation is unknown and must be estimated using the sample standard deviation, particularly when the sample size is small.

F. Hypothesis Testing Using Confidence Intervals

48. What is hypothesis testing?

Answer: Hypothesis testing is a statistical method used to make a decision about a population parameter based on sample data. It helps determine whether there is sufficient evidence to support or reject a particular claim.

49. What are the null hypothesis (H_0) and alternative hypothesis (H_1)?

Answer: The null hypothesis (H_0) is the initial assumption that there is no significant difference, effect, or relationship. The alternative hypothesis (H_1) is the statement that contradicts the null hypothesis and represents the claim being tested.

50. How can a confidence interval be used to make a decision about a hypothesis?

Answer: A confidence interval can be used to test a hypothesis by checking whether the hypothesized population parameter value lies within the interval. If the hypothesized value is inside the confidence interval, we fail to reject H_0 . If the hypothesized value is outside the confidence interval, we reject H_0 at the corresponding significance level.

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well- structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	